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Aerosol optical depth variation observed using sun-photometer over Indore

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Columnar aerosol optical depth measurements have been made over Indore (22.7° N, 75.9° E) during May 2001-March 2002 at four spectral bands centred around 400, 499, 668 and 858 nm using a hand-held sun-photometer. The aerosol optical depth (AOD) shows seasonal variation with high values in summer and low values in winter. The summer-increase is found to be due to the high wind speed producing larger amount of wind-derived dust particles. As the summer monsoon sets in at the end of June there is an appreciable decrease in the AOD values. The AOD values decrease further in winter and the decrease is more at higher wavelengths indicating that there is a general reduction in the number of bigger particles. Also during winter months the wind direction changed to southerly and south-easterly which brings air that is more rural to the measurement site. The amplitude of the observed high AOD values in summer and low during winter, is higher for longer wavelengths which shows that the coarse particles contribute more to the observed variation as compared to sub-micron particles. Comparison of the AOD values over Indore with that of Trivandrum and Visakhapatnam shows that the Indore values are comparable to that of Visakhapatnam, but much higher than the Trivandrum values indicating the influence of industrialization and higher population.

Key words: Aerosols, Aerosol optical depth, Sun-photometer

1 Introduction

Atmospheric aerosols exert both direct and indirect forcing on climate[1]. The direct effect is due to scattering and absorption of solar and terrestrial radiations by aerosols. They indirectly influence cloud radiative forcing[2] by modifying cloud microphysical properties. The magnitude of direct radiative forcing by aerosols at any location and time depends on the amount of aerosols present, their optical properties and underlying surface albedo. The major difference between the radiative forcing by greenhouse gases and aerosols is that the greenhouse gases forcing occurs both during day and nighttime and applicable for cloudy and cloud-free conditions[3]. Also, most of the greenhouse gases have very long lifetime compared to that of aerosols. Hence they are well mixed, and forcing is more or less uniform throughout the globe. However, tropospheric aerosols have very short lifetime compared to greenhouse gases and hence the aerosol forcing exhibits large spatial and temporal variations and has regional pattern[4]. Unlike the greenhouse gases, aerosols exert negative forcing and tend to reduce the global surface temperature by backscattering the incoming solar radiation to space[5]. However, aerosols like soot and desert dust can absorb lower wavelengths and re-emit at thermal wavelengths and contribute a small, but non-negligible amount to positive radiative forcing[6]. In addition, atmospheric aerosols are found to affect indirectly the amount, type and distribution of clouds. An increase in aerosol number concentration will result in an increase in the number of cloud condensation nuclei, which leads to clouds with higher number of smaller size cloud droplets[2]. An increase in the number of cloud droplets enhances multiple scattering of light within the clouds, which increases the optical depth and albedo of the cloud. Recent studies suggest that tropospheric aerosols contribute substantially to radiative forcing and anthropogenic sulphate aerosols and soot particles, in particular, have imposed a major perturbation to this forcing[7,8]. Therefore, it is very important to take aerosol effects into account while evaluating the anthropogenic influences on the past, present and future climate.

Aerosol optical depth (AOD) is an important parameter which tells about the amount of attenuation of

the incoming solar radiation due to aerosols and hence about the column burden of the aerosol loading. Daily monitoring of AOD can tell about the variation in column concentration of aerosols over the measurement site. The AOD is mainly due to both scattering and absorption of radiation and these processes depend on the size and composition of the particles as well as on the wavelength of the incoming solar radiation. Hence, monitoring of AOD at different wavelengths is useful to get additional information on the size distribution of particles, and study of its variation with season will help to identify the variation in the source strength of different particles emitted into the atmosphere.

In India there are multi-wavelength radiometer (MWR) stations located at Trivandrum, Mysore and Visakhapatnam, which are continuously monitoring the AOD since last one decade, but they all are at low latitudes in the southern part of India[9]. There is no station in the central India except the recently started station at Jodhpur. Indore (23°N, 76°E) is located at the central part of India and is a rapidly growing industrial city with more than three million population and more than 6,00,000 automobiles of different kinds. It is important to monitor the AOD over this new site and compare the data with other locations to delineate the relative contribution of anthropogenic activities in aerosol loading.

2 Experiment

In the present work a hand-held sun-photometer was used to measure the direct solar radiation intensities at five wavelength bands centred around 400, 499, 668, 858 and 960 nm with bandwidth between 13 and 29 nm. The total field of view of the instrument is kept around 8° using a baffle attached at the front of the photometer. Interference filters are used to select desired wavelength regions and a silicon photodiode is used as the light detector. The 960 nm channel is used for obtaining the columnar water vapour concentration, while the other four channels are used to obtain the aerosol optical depth. By unscrewing the baffle, filter assembly can be changed to take measurement at different wavelengths. Table 1 presents the characteristics of the interference filters used. The photometer is manually pointed towards the sun with the help of a sun-pointer attached to the instrument and observations of the direct solar radiation intensity in terms of voltage along with time are noted down.

Table 1 – Optical characteristics of different filters used in the hand-held sun-photometer for the columnar aerosol optical depth and water vapour measurements

Channel number	Central wavelength (λ_{max}) nm	FWHM (Bandwidth) nm	Transmission at λ_{max} %
1	400	19	22.9
2	499	13	46.0
3	668	13	40.7
4	858	29	02.9
5	960	20	44.0

Accuracy in time is very crucial for the estimation of the sun position at the time of observation and hence in the computation of the optical depth. So, before starting the observations the time measurement is set to IST within an accuracy of thirty seconds or better, using a GPS receiver. Observations were taken for all the filters at an average interval of 15 min starting from 0830 hrs IST to 1700 hrs IST on each clear and stable day from 29 May 2001 to 3 Mar. 2002 over Indore at two different locations. One of the sites is at the University campus and the other is at Tilak Nagar, a few kilometres north of the university campus (selected for the operational convenience). Both of these locations are at the eastern edge of city and do not make any difference in the transportation of aerosols and aerosol characteristics. Due to relatively larger number of cloudy days, only limited observations could be made in the months of July and August. Table 2 shows the total number of observations taken every month for each filter.

3 Theory

As solar radiation penetrates into the earth's atmosphere, it is absorbed and scattered by atmospheric gases and aerosols. For a monochromatic solar radiation with intensity I_o (λ) at the top of the atmosphere, and the transmitted intensity I (λ) that reaches the earth's surface, we can write (using Beer-Lambert relation),

$$I(\lambda) = I_o(\lambda) \exp(-\delta_\lambda m) \quad \ldots (1)$$

where, δ_λ is the total optical depth of the atmosphere and m the relative airmass (ratio of actual path to vertical path). In the atmosphere, since various constituents with different attenuation coefficients cause extinction and the processes occur independently of each other, the total optical depth due to all processes can be written as the sum of individual optical depths

Table 2 – Total number of observations made for each channel from May 2001 to March 2002 (Each observation is converted into aerosol optical depth, and daily and monthly mean values are derived from the individual observation).

Period	Channel				
	400 nm	499 nm	668 nm	858 nm	960 nm
May-01	21	21	21	21	21
June-01	108	133	130	97	97
July-01	25	24	24	24	24
Aug.-01	50	50	48	42	42
Sep.-01	131	131	130	126	126
Oct.-01	175	173	170	166	166
Nov.-01	279	279	279	277	277
Dec.-01	135	135	134	132	132
Jan.-02	303	304	303	297	297
Feb.-02	183	183	182	181	181
Mar.-02	37	37	37	37	37
Total	1447	1470	1458	1400	1400

The measured solar radiation intensity is used to calculate the total optical depth of the atmosphere using the following relation, which takes care of the effect of earth-sun distance variations over I_0.

$$\delta = -(1/m)\{\ln(I/I_o) - 2\ln(r_0/r)\} \quad \ldots (2)$$

where, I is instantaneous solar radiation intensity measured in volt, I_0 the equivalent solar radiation intensity at the top of the atmosphere derived using Langley plot technique, r_0 the sun-earth mean distance when I_0 value is evaluated and r the instantaneous sun-earth distance at the time of measurement of I. The relative air mass (m) is the term used to account for the relative change in the atmospheric path length traversed by the solar radiation with respect to the zenith. In the present work air mass is calculated using the following relation given by Young[10]:

$$m = \frac{1.002432\cos^2(z) + 0.148386\cos(z) + 0.096467}{\cos^3(z) + 0.149864\cos^2(z) + 0.0102963\cos(z) + 0.000303978} \quad \ldots (3)$$

This relation takes care of atmospheric refraction and earth's curvature effect and provides better accuracy even at higher zenith angles. The solar zenith angle (z) is computed for the time of observation using the relation[11],

$$\cos(z) = \cos(d)\cos(\phi)\cos(h) + \sin(d)\sin(\phi) \quad \ldots (4)$$

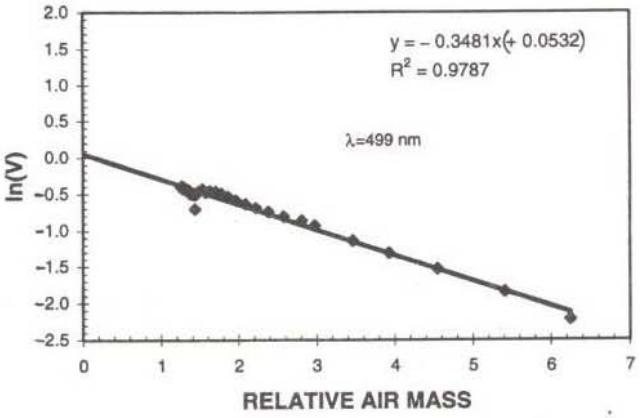


Fig. 1—Langley plot for filter 499 nm attempted over Indore on 9 Feb. 2002

where, d is the sun's declination, ϕ the latitude of the location and h the hour angle.

The I_0 (λ) values are obtained using Langley plot technique. This gives I_0 (λ) for a particular photometer and for a particular channel. The voltage I_0 (λ) obtained for zero air mass is a constant for a particular photometer, if the response of the instrument is constant with time and if the value is corrected for mean solar distance. Figure 1 shows a sample Langley plot for the 499 nm channel used in the experiment over Indore on 9 Feb. 2002. The amount of solar radiation reaching the earth is inversely proportional to the square of its distance from the sun and hence an accurate value of sun-earth distance is required. The sun-earth distance r for any day of any year is known with considerable accuracy.

In equation 3 , typographic error is there. Correct constant is 0.0096467

The total atmospheric optical depth can be written as:

$$\delta_T = \delta_{RS} + \delta_{ma} + \delta_{aerosol} \quad \ldots (5)$$

where, δ_{RS} is the Rayleigh scattering optical depth, $\delta_{aerosol}$ the aerosol optical depth, δ_{ma} the optical depth due to molecular absorption by gases like water vapour, ozone, etc. The aerosol optical depth is obtained by subtracting δ_{RS} and δ_{ma} from δ_T.

Figure 2 shows the relative contributions of various atmospheric constituents to the total optical depth for a typical atmosphere. The optical depth due to Rayleigh scattering is calculated by multiplying the Rayleigh scattering cross-section [σ_{eff} (λ)] for the effective wavelength of a particular channel and the integrated columnar number density (N) of air. For the column integrated air number density, tropical model atmosphere[12] is used, where N = 2.153E+25 cm^{-2}. Since each filter used in the experiment has certain bandwidth on both sides of λ_{max}, an effective δ_{RS} is to be calculated using the relation

$$\sigma_{eff}(\lambda) = \frac{\int_{\lambda_1}^{\lambda_2} \sigma_i(\lambda) FT(\lambda) d\lambda}{\int_{\lambda_1}^{\lambda_2} FT(\lambda) d\lambda} \quad \ldots (6)$$

where, $\sigma_{eff}(\lambda)$ is the effective Rayleigh scattering cross-section in cm^2 for a particular filter bandwidth, $\sigma_i(\lambda)$ is the Rayleigh scattering cross-section in cm^2 for a single wavelength and FT(λ) the filter transmission in percentage for the wavelength range from λ_1 to λ_2. The parameter $\sigma(\lambda)$ is computed using Nicolet's[13] equation. The accuracy of Nicolet's formula is better than ± 0.5%. Filter transmission characteristics for each filter is measured in the laboratory using a double-beam spectrometer. The calculated values of δ_{RS} are shown in Table 3 that are used in the derivation of AOD.

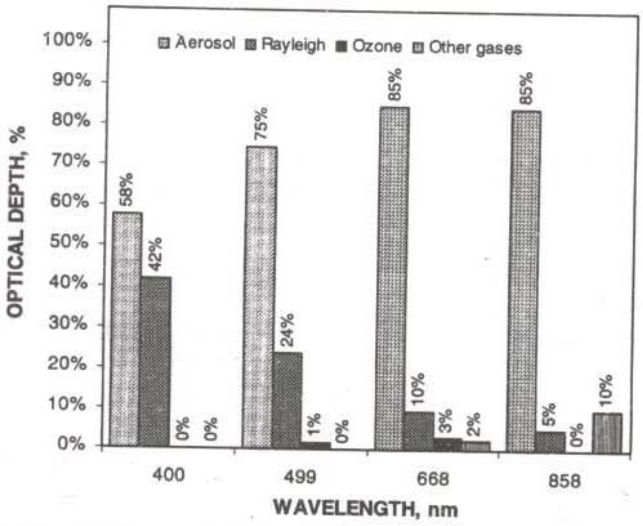


Fig. 2—Relative contributions to total optical depth from various constituents for a typical urban atmosphere

Optical depth due to ozone absorption is calculated using the total ozone maping spectrometer (TOMS)-derived monthly averaged total column ozone amount over the region, taken from the web site[14]. Ozone optical depth for a particular channel has been calculated following exactly the same method as used for computing the Rayleigh scattering optical depth. The only change, however, is the use of ozone absorption cross-section[15] instead of the scattering cross-section. Figure 3 shows the total ozone variation over Indore throughout the period of observation. The column ozone amount is found to peak in June with a value around 290 DU and shows a minimum of 230-240 DU in November-December. From May to July, the value remains more or less constant. This variation in the column amount of ozone over the measurement site is accounted for while computing the optical depth due to ozone absorption that is subtracted from the total optical depth for the determination of the aerosol optical depth. To calculate optical depth due to other atmospheric gases, default concentrations for a tropical model atmosphere[12] is used. Important of them is the water vapour concentration. However, optical depth due to water vapour is zero for 400 nm and 499 nm filters, while it is less than 10% of total optical depth for 858 nm filter for default concentrations taken for a tropical atmosphere. Later, it has been shown that the observed water vapour concentration is much less than the model atmosphere. Table 3 shows the optical depths due to ozone and water vapour absorptions and Rayleigh scattering for the model atmosphere.

Table 3 – Optical depth computed for gaseous absorptions by ozone and water vapour and Rayleigh scattering for a tropical model atmosphere[12]

Channel number	Central wavelength (λ_{max}) nm	δ_{ozone}	δ_{water}	δ_{RS}
1	400	1.374E-04	0.000E+00	0.366
2	499	9.253E-03	0.000E+00	0.147
3	668	1.433E-02	9.217E-03	0.044
4	858	0.000E+00	3.897E-02	0.017

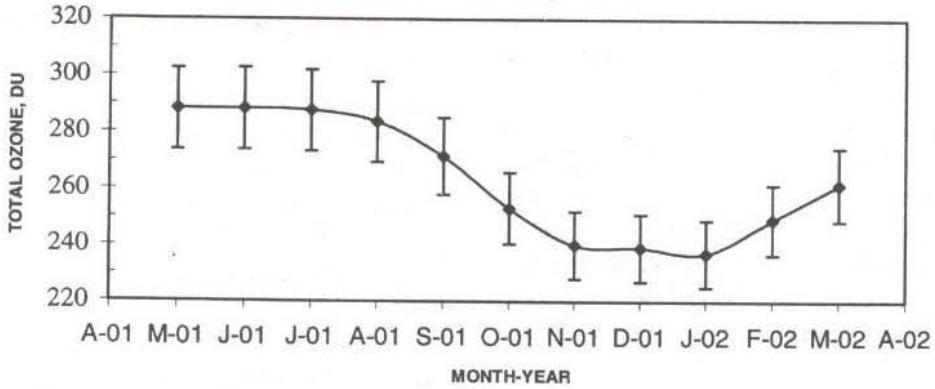


Fig. 3—Total ozone maping spectrometer-derived monthly mean total columnar ozone concentration in Dobson Unit (DU) over Indore [Vertical bars represent ±1 standard deviation of the mean.]

4 Meteorological conditions over Indore during the observation period

The variation in monthly mean wind speed and direction starting from May 2001 to December 2001 is shown in Fig. 4. The wind speed is maximum with a value of about 10-11 m/s during the months of May and June, whereas it is minimum with a value of about 2-3 m/s in the month of December. In the month of June, on some occasions, the wind speed reached up to 23 m/s. It starts to decrease from June and decreases continuously till December. Moderate (6-8 m/s) to high (>8 m/s) wind patterns were observed from May to August and winds were generally calm (0-5 m/s) from September to December. Monthly mean wind speed showed an almost constant standard deviation of about 3.5 m/s at all months. The wind direction during the months of May to September is observed to be westerly between 285° and 240°. During this period a gradual change is observed in the wind direction from westerly to southerly. In the month of October, the wind direction becomes almost southerly (180°). From October to November the direction changed from southerly to south-easterly (135°) and in December it became southerly again. The computed monthly mean wind direction has a maximum standard deviation of about 70° in October, showing maximum variability and a minimum of 16° in May.

The variation in the observed monthly mean relative humidity (RH) and total rainfall from May to December 2001 is shown in Fig. 5[(a) and (b)]. The vertical bars show ±1 standard deviation of the mean. The monthly mean RH shows the peak value of about 90% in the month of July. Daily data show that humidity reached 98% and above on a few days in the months June to September, when the summer monsoon is active over the location. Monthly mean RH value remains almost same in July and August and starts decreasing after August and continuously decreases till November. In December, it shows a small increment by about 5%. Standard deviation shows a maximum value of about 20% in the month of October and minimum during July-August with a value of about 8-10%.

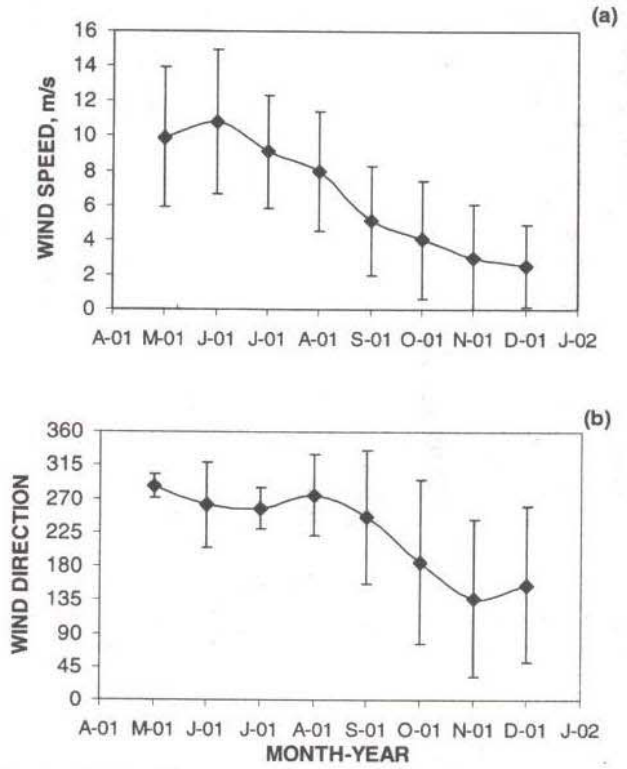


Fig. 4—(a) Monthly mean wind speed and (b) Wind direction, measured over Indore for the year 2001 [The vertical bars in the plots show the standard deviation (± σ) of the mean for that month.]

Total rainfall received in each month is shown in Fig. 5(b) as bar chart. The maximum rainfall occurred

during the month of June in the year 2001. It is to be noted that the rainfall occurred during the end of the month and on these days AOD observations were not taken, and the high AOD value obtained in June, prior to the rain, should not be confused with the high rain. July and August months had less rainfall and in September it was very less with a value of about 10 mm. May, October, November and December months had also one or two rainy days. Due to instrumental failure meteorological data could not be collected during January-March 2002.

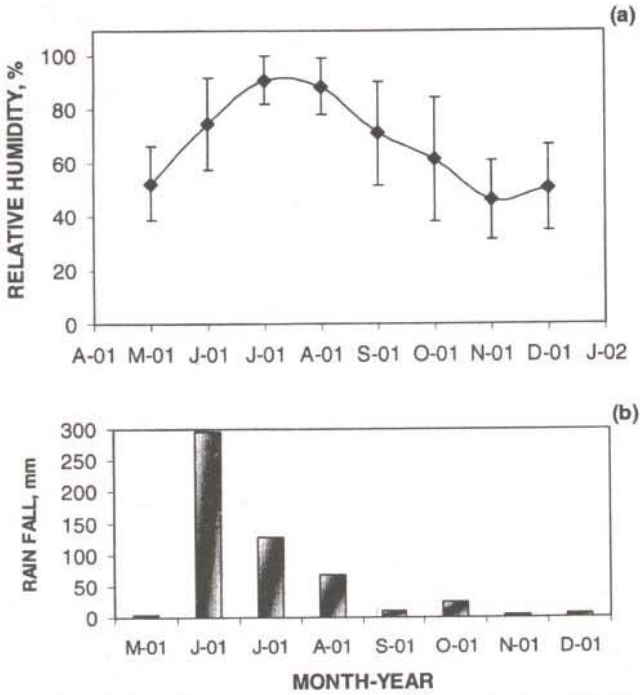


Fig. 5—(a) Monthly mean relative humidity variation with standard deviation and (b) Total rainfall observed over Indore during May-December 2001

5 Estimation of column water vapour concentration

Column water vapour concentration is computed using the 960 nm channel data during the measurement period. The interference filter used in this channel has a peak transmission of about 44% at 960 nm and has a bandwidth of about 20 nm. The filter transmission characteristics overlap molecular absorption band of water vapour. For average condition, percentage contribution of water vapour absorption optical depth to the total optical depth is about 50% for this channel. To obtain the optical depth by water vapour, Rayleigh and aerosol contributions to the total optical depth need to be subtracted. Rayleigh optical depth is calculated in the same way as for other filters. Aerosol optical depth at 960 nm is obtained by using the power law,

$$\delta = \beta \lambda^{-\alpha} \qquad \ldots (7)$$

where, the parameters α and β are calculated for each day by least square fitting the daily AOD values from 400 nm to 850 nm. Using the Santa Barbara Discrete Ordinate Atmospheric Radiative Transfer Model (SBDART)[16] the following relationship between columnar water vapour and water vapour optical depth for the given filter is obtained.

$$\text{Water vapour concentration (g/cm}^2) = -1.6434\,\delta^3 + 6.8224\,\delta^2 + 1.4488\,\delta - 0.0899 \qquad \ldots (8)$$

The derived column water vapour concentration is shown in Fig. 6. The daily mean surface level water vapour concentration in g/cm^3 derived from RH and temperature values is also shown in Fig. 6. Both the

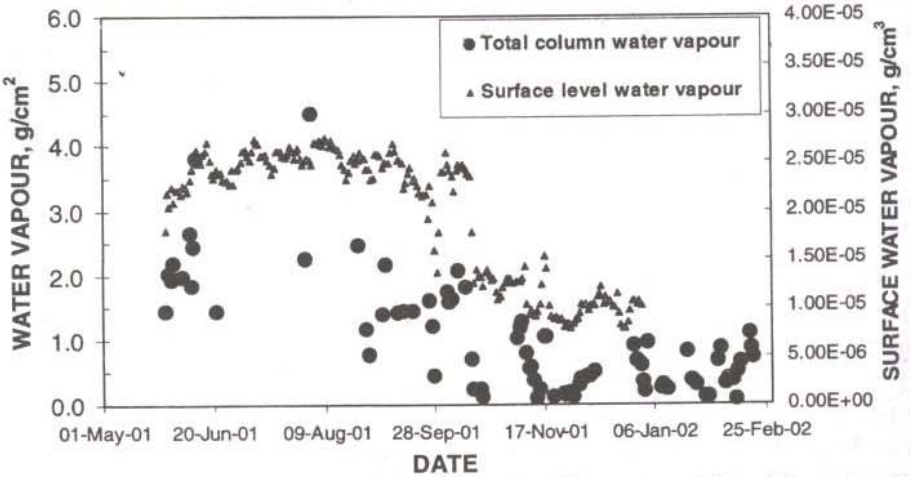


Fig. 6—Daily mean column water vapour concentration estimated from the 960-nm channel data [The surface level water amount derived from relative humidity and temperature measured for the corresponding days are also plotted for comparison.]

column and surface level water amount show a more or less similar seasonal trend. The maximum total column water vapour value observed was 4.5 g/cm^2 during August. During winter months the value remained always less than 1 g/cm^2. Here, it is to be noted that average water vapour concentration suggested in US 1962 model atmosphere[12] for tropical region is 4.2 g/cm^2, which is, however, true only for the summer monsoon period and not for winter months over Indore.

6 Results and discussion

Monthly averaged values of AOD for the entire period of observation (29 May 2001-3 Mar. 2002) and for all the four wavelengths are shown in Fig. 7. The AOD values show more or less similar behaviour at all the wavelengths during the entire period of observation. In May, the AOD values for different wavelengths range from 0.3 to 0.4. The increase in June from 0.4 to 0.6 could be caused mainly due to increased dust input into the atmosphere by strong monsoon winds observed in that month. Possibility of cloud processing of aerosol[17] resulting in an increase in optical depth cannot be ruled out. It has been shown[17] that cloud water can serve as the reacting medium for a series of aqueous phase reactions such as the conversion of sulphur dioxide to sulphate particles, which result in an overall increase in the particles size and amount. It may be noted that the highest AOD value for the entire observation period is recorded in the month of June. The AOD values are found to decrease as the monsoon rain starts in June end, which is also reflected in the July AOD values. July values are found very similar to May values. This signifies the fact that wind-blown dust particle are more effectively removed by rain wash than the anthropogenically produced urban aerosol, as the later is much smaller in size compared to the former. In July and August, when the S-W monsoon is active over the region, the AOD values are low, lying in the range of 0.3-0.4. In September, the monthly mean AOD shows a rise compared to July and August. The monsoon rainfall has also decreased in September. The total rainfall recorded at Indore in September 2001 was only 10 mm compared to 129 mm and 69 mm in the months of July and August 2001, respectively. From September to October, AOD decreases and remains more or less steady till December with small monthly variation. From January to February, a small increment of 0.04 (on an average) for all the four wavelengths was observed which again fell by 0.11 in the month of March.

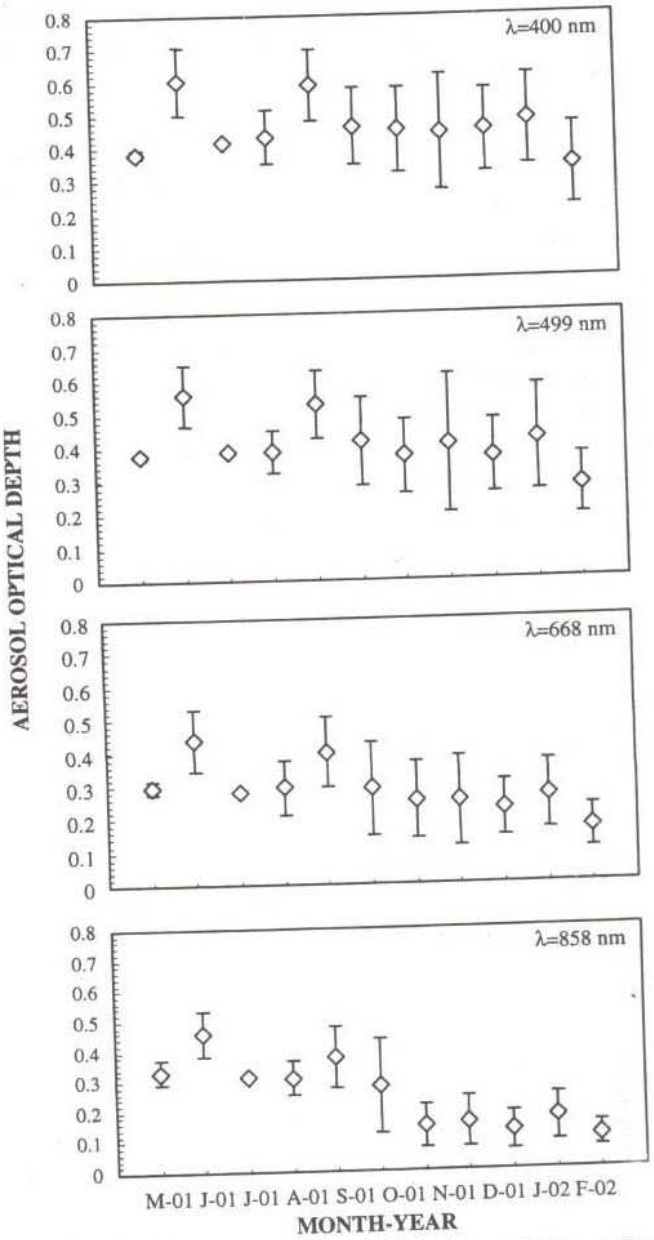


Fig. 7—Monthly mean aerosol optical depth over Indore at four wavelengths for the period (May 2001-March 2002) of observation [The vertical bar shows standard deviation (± σ) of the mean. In May and July months the standard deviation is very low and may not be visible in some cases.]

In order to study the overall variation in the AOD values the monthly mean data are grouped into four seasons, viz. summer, rainy, winter and post-winter (spring) seasons. Months of May and June represent summer; July, August and September represent rainy; October, November and December represent winter and January, February and March months represent

post-winter, respectively. The AOD values grouped according to the above seasons for all the 4 wavelengths with standard deviations are shown in Fig. 8. The seasonal variation in aerosol optical depth shows more or less similar pattern for all the wavelengths. The peak aerosol optical depth values are observed in summer and minimum in winter and spring seasons. The higher AOD values observed during summer are due to the combined effects of high wind speed and a favourable wind direction. It may be noted that during the summer and rainy seasons the wind direction is mainly westerly and, with respect to the present observation site, these winds are from the thickly populated as well as industrial area of the Indore city, which resulted in bringing in the large amount of dust and anthropogenically produced particles to the measurement site. Particularly the boundary layer aerosols, which contribute to more than 80% of the observed columnar content, are highly influenced by the local aerosol sources. In the rainy season, though the winds continue to be from the polluted region, the rain-wash could reduce the aerosol loading. The AOD values decrease in winter, which is mainly because of reduction in bigger size particle. This is evident from the fact that the Angstrom's exponent α in Eq. (7) is found to be around 0.3-0.5 during May-July and around 1.5-1.7 during November-January. Lower value of α means dominance of bigger size particle in the aerosol size distribution and vice versa. This is due to the reduction of the wind-derived dust particles into the atmosphere, as the wind speed is found to be decreasing (Fig. 4) continuously from a maximum of about 10 m/s to a low of about 3 m/s in the winter months. Also during winter months the wind direction changed to southerly and south-easterly which brings air that is more rural to the measurement site. In spring, a slight increase in AOD is observed for lower wavelengths, whereas the AOD at higher wavelengths show a continuous decrease. Thus, the observed seasonal variation should be an annual feature with high values in summer and low during winter. The amplitude of the variations is higher for higher wavelength showing that the coarse particles contribute more to the observed variation compared to smaller particles.

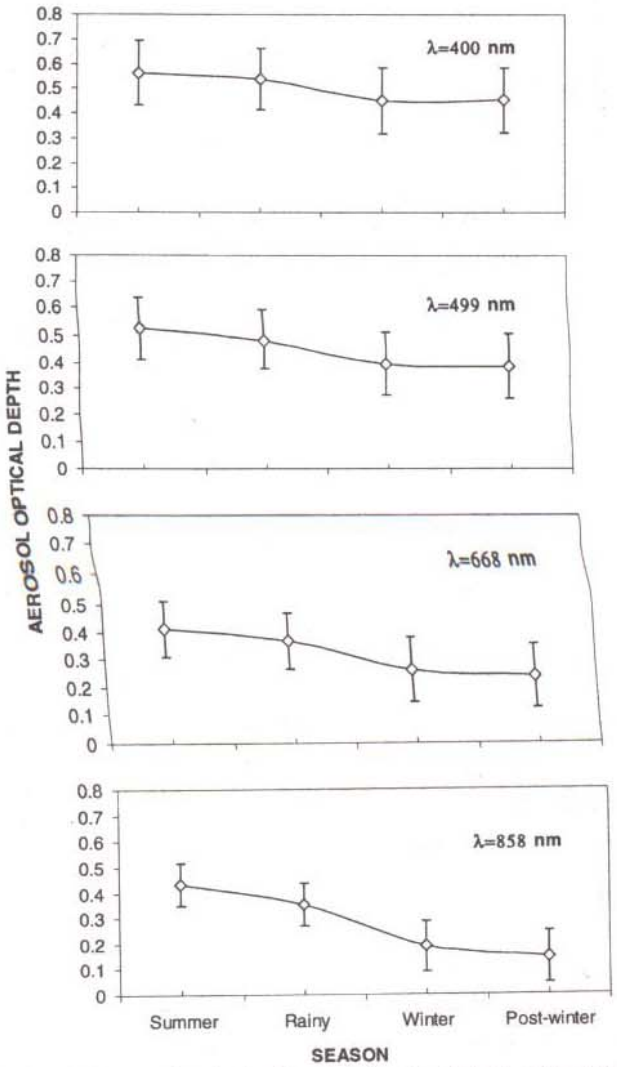


Fig. 8—Seasonal variation in aerosol optical depth with standard deviation at four wavelengths over Indore [Seasons are classified as summer (May-June), rainy (July-September), winter (October-December) and post-winter (January-March).]

7 Comparison of AOD with data from other stations

Krishna Moorthy[18] has presented the climatology of aerosol optical depths and long-term trends for the MWR stations, viz. Trivandrum, Mysore and Visakhapatnam. The AOD values are available till the beginning of 2001 for Trivandrum and Visakhapatnam. The Trivandrum AOD values for the 400 and 500 nm channels at the beginning of 2001 are about 0.35. The average AOD value for Indore for the year 2001 for these two channels are 0.47±0.08 and 0.43±0.07. In the case of Visakhapatnam, the AOD values at the beginning of 2001 are about 0.48 and 0.42 for the 400 and 500 nm channels, respectively. The comparison shows that the Indore values are much higher than the Trivandrum values by about 31% and 19% for the above two channels but comparable to the Visakhapatnam values. Trivandrum city is less industrialized compared to Visakhapatnam and Indore, showing the influence of the anthropogenically produced aerosols in the later two cities.

8 Summary and conclusions

Regular observations of aerosol optical depth (AOD) at four wavelength bands and total columnar water vapour content were taken during the period 29 May 2001-3 Mar. 2002 over Indore. The AOD values show seasonal pattern with high values in summer and low values in winter. The summer-increase is found to be the combined effect of high wind speed observed in the month of June producing larger amount of wind-derived dust particles and favourable wind direction which brought pollutants from the thickly populated city region to the measurement site. As the summer monsoon sets in at the end of June there is an appreciable decrease in the AOD values. The AOD values decrease further in winter and the decrease is more at higher wavelengths indicating that there is a general reduction in the number of bigger size particles. This is mainly due to the reduction of the wind-derived dust particles into the atmosphere, as the wind speed is found decreasing from summer to winter. Also during winter months the wind direction changed to southerly and south-easterly which brings air that is more rural to the measurement site. The amplitude of the observed variation, with high AOD values in summer and low during winter, is higher for higher wavelength which shows that the coarse particles contribute more to the observed variation compared to sub-micron particles. Comparison of the AOD values over Indore with that of Trivandrum and Visakhapatnam shows that the Indore and Visakhapatnam values are comparable and much higher than the Trivandrum values. Indore and Visakhapatnam have more industries and higher population compared to Trivandrum and hence the anthropogenic effect is responsible for the higher AOD observed in the former two cities. For the development of aerosol climatology over a region, long-term data are essential. Due to logistics problem, only limited data could be collected over Indore. However, attempts are being made to make a permanent station in the central Indian region that will help in evolving an aerosol climatology for India along with the MWR data.

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